chromosome consists of a single, continuous DNA double helix; that is, each chromosome is one long, double-stranded DNA molecule, and the nuclear genome consists, therefore, of 46 linear DNA molecules, totaling more than 6 billion nucleotide pairs (see Fig. 2-1).

Chromosomes are not naked DNA double helices, however. Within each cell, the genome is packaged as **chromatin,** in which genomic DNA is complexed with several classes of specialized proteins. Except during cell division, chromatin is distributed throughout the nucleus and is relatively homogeneous in appearance under the microscope. When a cell divides, however, its genome condenses to appear as microscopically visible chromosomes. Chromosomes are thus visible as discrete structures only in dividing cells, although they retain their integrity between cell divisions.

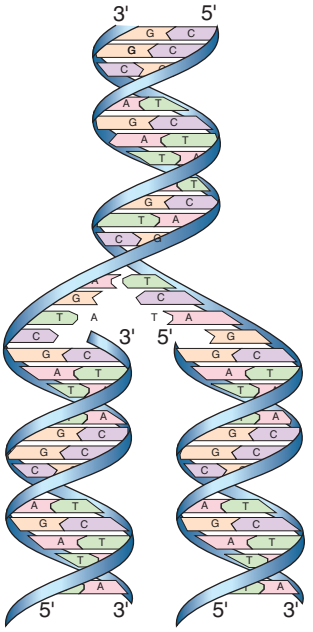


Figure 2-4 Replication of a DNA double helix, resulting in two identical daughter molecules, each composed of one parental strand and one newly synthesized strand.

The DNA molecule of a chromosome exists in chromatin as a complex with a family of basic chromosomal proteins called *histones*. This fundamental unit interacts with a heterogeneous group of nonhistone proteins, which are involved in establishing a proper spatial and functional environment to ensure normal chromosome behavior and appropriate gene expression.

Five major types of histones play a critical role in the proper packaging of chromatin. Two copies each of the four core histones H2A, H2B, H3, and H4 constitute an octamer, around which a segment of DNA double helix winds, like thread around a spool (Fig. 2-5). Approximately 140 base pairs (bp) of DNA are associated with each histone core, making just under two turns around the octamer. After a short (20- to 60-bp) "spacer" segment of DNA, the next core DNA complex forms, and so on, giving chromatin the appearance of beads on a string. Each complex of DNA with core histones is called a **nucleosome** (see Fig. 2-5), which is the basic structural unit of chromatin, and each of the 46 human chromosomes contains several hundred thousand to well over a million nucleosomes. A fifth histone, H1, appears to bind to DNA at the edge of each nucleosome, in the internucleosomal spacer region. The amount of

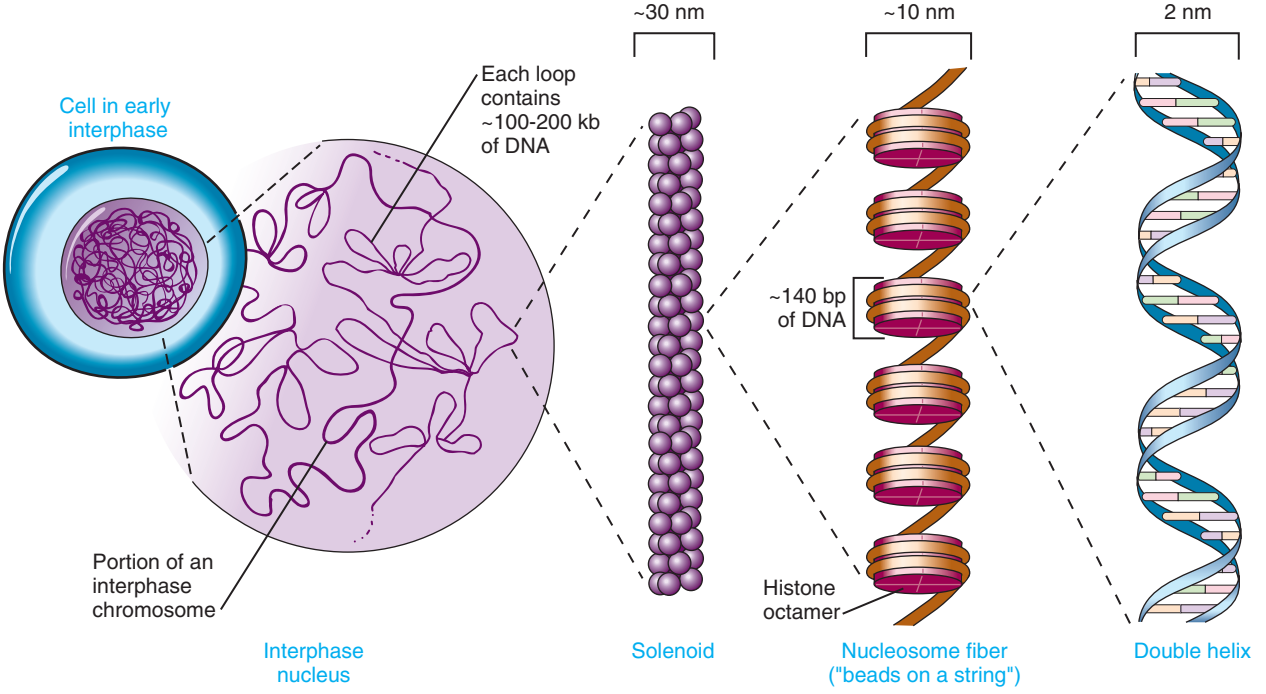


Figure 2-5 Hierarchical levels of chromatin packaging in a human chromosome.